



Separable Differential Equations

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Separate variables in a first-order differential equation
- Integrate both sides to find the general solution
- Apply initial conditions to determine the particular solution

Solve each separable differential equation using the initial condition to find the particular solution for y .

1. Find the equation of the curve passing through (0, 7) with the given slope.

$$\frac{dy}{dx} = 4x^3y$$

Answer: _____

2. Solve the separable differential equation with the given initial condition.

$$\frac{dy}{dx} = 2xy, y(0) = 3$$

Answer: _____

3. Find the particular solution to the differential equation.

$$\frac{dy}{dx} = 6x^2y, y(0) = 5$$

Answer: _____

4. Solve the differential equation with the initial point given.

$$\frac{dy}{dx} = \frac{y}{x}, y(1) = 4$$

Answer: _____

5. Find y as a function of x .

$$\frac{dy}{dx} = 3x^2y, y(0) = 2$$

Answer: _____

6. Determine the curve passing through the given point.

$$\frac{dy}{dx} = -2xy, y(0) = 10$$

Answer: _____



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7. Solve the separable equation using the initial condition.

$$\frac{dy}{dx} = 5y, \quad y(0) = 1$$

Answer: _____

8. Find the particular solution of the differential equation.

$$\frac{dy}{dx} = 8x^3y, \quad y(0) = 6$$

Answer: _____

9. Solve the differential equation with the given condition.

$$\frac{dy}{dx} = x^2y, \quad y(0) = 4$$

Answer: _____

10. Find the function y satisfying the equation.

$$\frac{dy}{dx} = \frac{4x^3}{y}, \quad y(0) = 2$$

Answer: _____





Emphasize the three-step process: separate variables, integrate both sides, then apply the initial condition to solve for the constant C .

Solutions

1. Find the equation of the curve passing through $(0, 7)$ with the given slope.

$$\frac{dy}{dx} = 4x^3y$$

- Separate variables to get dy over y equals $4x$ cubed dx .
- Integrate both sides to get $\ln$ of absolute value of y equals x to the fourth plus C .
- Substitute x equals 0 and y equals 7 to find C equals $\ln 7$.
- Exponentiate both sides and simplify to get y equals 7 times e to the x to the fourth.

Answer: $y = 7e^{x^4}$

2. Solve the separable differential equation with the given initial condition.

$$\frac{dy}{dx} = 2xy, y(0) = 3$$

- Separate variables to get dy over y equals $2x$ dx .
- Integrate both sides to obtain $\ln$ of absolute value of y equals x squared plus C .
- Apply the initial condition y of 0 equals 3 to find C equals $\ln 3$.
- Exponentiate both sides to get y equals 3 times e to the x squared.

Answer: $y = 3e^{x^2}$

3. Find the particular solution to the differential equation.

$$\frac{dy}{dx} = 6x^2y, y(0) = 5$$

- Separate variables to get dy over y equals $6x$ squared dx .
- Integrate both sides to get $\ln$ of absolute value of y equals $2x$ cubed plus C .
- Use y of 0 equals 5 to determine C equals $\ln 5$.
- Exponentiate both sides to obtain y equals 5 times e to the $2x$ cubed.

Answer: $y = 5e^{2x^3}$

4. Solve the differential equation with the initial point given.

$$\frac{dy}{dx} = \frac{y}{x}, y(1) = 4$$

- Separate variables to obtain dy over y equals dx over x .
- Integrate both sides to get $\ln$ of absolute value of y equals $\ln$ of absolute value of x plus C .
- Substitute x equals 1 and y equals 4 to find C equals $\ln 4$.
- Exponentiate and simplify to get y equals $4x$.

Answer: $y = 4x$



5. Find y as a function of x .

$$\frac{dy}{dx} = 3x^2y, \quad y(0) = 2$$

- Separate variables to get dy over y equals $3x$ squared dx .
- Integrate both sides to obtain $\ln$ of absolute value of y equals x cubed plus C .
- Apply y of 0 equals 2 to determine C equals $\ln 2$.
- Exponentiate both sides to get y equals 2 times e to the x cubed.

Answer: $y = 2e^{x^3}$

6. Determine the curve passing through the given point.

$$\frac{dy}{dx} = -2xy, \quad y(0) = 10$$

- Separate variables to get dy over y equals negative $2x$ dx .
- Integrate both sides to obtain $\ln$ of absolute value of y equals negative x squared plus C .
- Use y of 0 equals 10 to find C equals $\ln 10$.
- Exponentiate both sides to get y equals 10 times e to the negative x squared.

Answer: $y = 10e^{-x^2}$

7. Solve the separable equation using the initial condition.

$$\frac{dy}{dx} = 5y, \quad y(0) = 1$$

- Separate variables to obtain dy over y equals 5 dx .
- Integrate both sides to get $\ln$ of absolute value of y equals $5x$ plus C .
- Apply y of 0 equals 1 to find C equals 0 .
- Exponentiate both sides to get y equals e to the $5x$.

Answer: $y = e^{5x}$

8. Find the particular solution of the differential equation.

$$\frac{dy}{dx} = 8x^3y, \quad y(0) = 6$$

- Separate variables to obtain dy over y equals $8x$ cubed dx .
- Integrate both sides to get $\ln$ of absolute value of y equals $2x$ to the fourth plus C .
- Use y of 0 equals 6 to determine C equals $\ln 6$.
- Exponentiate both sides to obtain y equals 6 times e to the $2x$ to the fourth.

Answer: $y = 6e^{2x^4}$

9. Solve the differential equation with the given condition.

$$\frac{dy}{dx} = x^2y, \quad y(0) = 4$$

- Separate variables to obtain dy over y equals x squared dx .
- Integrate both sides to get $\ln$ of absolute value of y equals x cubed over 3 plus C .
- Apply y of 0 equals 4 to determine C equals $\ln 4$.
- Exponentiate both sides to get y equals 4 times e to the x cubed over 3 .

Answer: $y = 4e^{\frac{x^3}{3}}$



10. Find the function y satisfying the equation.

$$\frac{dy}{dx} = \frac{4x^3}{y}, \quad y(0) = 2$$

→ Separate variables to obtain $y \, dy$ equals $4x$ cubed dx .

→ Integrate both sides to get y squared over 2 equals x to the fourth plus C .

→ Apply y of 0 equals 2 to find C equals 2.

→ Multiply by 2 and take the square root to get y equals the square root of $2x$ to the fourth plus 4.

Answer: $y = \sqrt{2x^4 + 4}$

